SR series/TS2000

Industrial Robot

INSTRUCTION MANUAL

WAFER TRANSFER ROBOT

TRANSPORTATION & INSTALLATION

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before installing and operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

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Shibaura Machine

TOKYO, JAPAN

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PREFACE

This manual describes how to unpack and install the robot and controller. Specifically, it describes how to unpack the shipment containing the equipment, how to connect wiring and air piping, and how to attach tools. Be sure to look through this manual before unpacking the shipment.

Before beginning the work according to this manual, read through the Setup Manual and Safety Manual so that you can learn the names and functions of various components and the safety measures.

This manual is divided into the following four (4) sections.

Section 1: Transportation
This section describes how to remove the robot and controller from their boxes and how to carry them to the installation site.

Section 2: Installation
This section discusses the equipment installation environment, space requirements and how to install the equipment.

Section 3: System Connections
This section describes how to connect the robot, controller and peripheral equipment.

Section 4: Tool Interface
This section describes how to connect the tool to the robot arm and how to connect pipes and wires to the tool. This section also discusses maximum permissible loads of the tool.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Be sure to read through the safety instruction manual provided separately before starting actual work to assure the safe work covering from the robot installation to operation.

[Installation and transportation]

Be sure to observe the following items to use the robot safely.

DANGER	
 Prohibited	• DO NOT install or operate if any parts are damaged or missing. Otherwise, electric shocks, fires or faults could be caused. • DO NOT install the robot where it may be exposed to splash of liquids such as water. Otherwise, electric shocks, fires or faults could be caused. • DO NOT place the robot near the combustible. Otherwise, fires could be caused if the material ignites due to a fault, etc.
 Mandatory	• Be sure to secure the robot with the attached clamps before carrying the robot. Otherwise, you could be involved in injuries if the arm moves down when the robot is lifted. • Connect the robot only after installation. If wiring is performed before the robot installation, you could be involved in electric shocks or injuries. • Be sure to use the power supply voltage and power capacity designated by SHIBAURA MACHINE. Otherwise, the equipment could be damaged or catch fires. • Be sure to use the designated power cable. Otherwise, fires or faults could be caused.
 Always ground	• Completely connect the grounding cable. Otherwise, electric shocks or fires could be caused if a fault or leak current occurs, in addition to malfunction by noise.

CAUTION	
 Prohibited	• NEVER lift the robot arm during the transport. Otherwise, an excessive force is exerted on the mechanical part of the robot, resulting in faults. • For the controller, keep an ample space for air vent. Otherwise, the controller will be heated, causing faults.
 Mandatory	• When lifting up the robot, do so carefully. The robot will tilt slightly. If it is lifted suddenly, it is very dangerous. • When storing the robot, completely clamp the base. If placed directly on the floor, the robot will be unstable and fall over.

Table of Contents

Page

Section 1 Transportation 1–1
1.1 Unpacking 1–1
1.2 Transportation 1–2
1.2.1 Mass and External Dimensions 1–2
1.2.2 Transporting the Robot 1–2
1.2.3 Transporting the Controller 1–4
1.3 Storage 1–4
1.3.1 Cautions on Robot Storage 1–4
1.3.2 Cautions on Robot Controller Storage 1–4

Section 2 Installation 2–1
2.1 Installation Environment 2–1
2.2 Installing the Robot 2–2
2.2.1 External Dimensions 2–2
2.2.2 Working Envelope 2–2
2.2.3 Robot's Origin Posture and Coordinate System 2–4
2.2.4 Robot Installation Procedures 2–5
2.3 Installing the Controller 2–6
2.3.1 External Dimensions 2–6
2.3.2 Rack Mounting Dimension 2–7
2.3.3 Cautions on Rack Mounting 2–7
2.3.4 Cautions on Direct Installation 2–10
2.4 Cautions on Handling of Teach Pendant 2–11
2.5 Safety Measures 2–12
2.6 Position Detector Backup Batteries 2–13

Page

Section 3 System Connections 3–1
3.1 Cable Wiring 3–1
3.1.1 Connector Arrangement on the Controller 3–1
3.1.2 Connecting Power Cable 3–2
3.1.3 Connecting Motor Cable 3–3
3.1.4 Connecting Encoder Cable 3–3
3.1.5 Connecting Robot Control Signal Cable 3–3
3.1.6 Inserting and Removing Connectors 3–3
3.1.7 List of Connector Terminal Arrangement Examples 3–5
3.2 Controller Connector Signals 3–7
3.2.1 Connector Signal Connection Diagrams 3–7
3.2.2 Jumpers for Safety Related Signals 3–7
3.3 Separating Control Panel from Controller 3–9
3.3.1 Removing Control Panel 3–9
3.3.2 Cable between Controller and Control Panel 3–10
3.3.3 Control Panel Mounting Dimensions 3–10
3.3.4 Mounting Dummy Panel on Controller 3–11
3.3.5 Dimensions when Separating Control Panel 3–12

Section 4 Hand Interface 4–1
4.1 Hand Wiring 4–1

Section 1 Transportation

1.1 Unpacking

The robot and controller are shipped separately in wooden frames and corrugated cardboards. The packaging state of the controller is shown in Fig. 1.1.
The teach pendant and other accessories including instruction manuals are packed separately.
Open the packages in a location easily accessible to where the equipment is to be installed. At this time, take careful precautions not to damage the robot and controller. After unpacking, make sure that all accessories are present and that all equipment have not been damaged during transport.

! DANGER

- If any parts of the equipment are found defective or damaged, or if some accessories are missing when the robot has reached your office, DO NOT install or operate the equipment. Otherwise, the equipment could malfunction. Contact us immediately at our sales agent in your territory.
- After unpacking, dispose of the wooden frames, corrugated cardboards, polyethylene shipping bags and cushion materials according to your regulations.

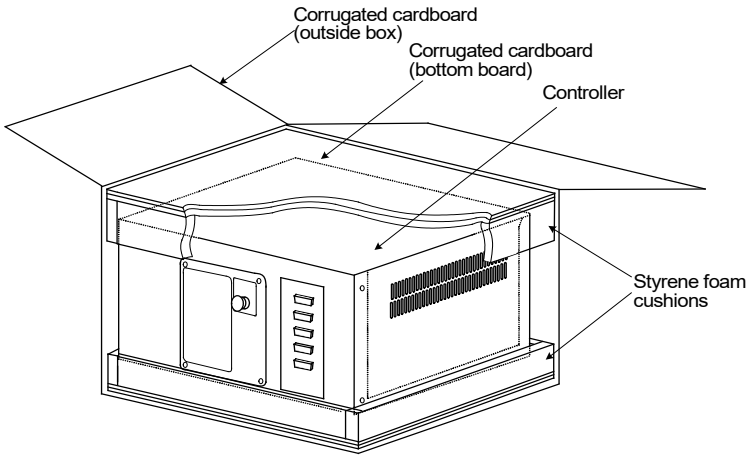


Fig. 1.1 Controller packaging state

1.2 Transportation

Carefully move the robot and controller to avoid excessive impact (9.8 m/s^2 or over) or vibration from being exerted on them. If the equipment are subject to vibration over a long period, be sure to completely tighten all nuts and screws.
When relocating the equipment at another place, put the equipment back into the wooden frames and corrugated cardboards, then reposition the cushions as they are.

1.2.1 Mass and External Dimensions

The mass of controller TS2000 is 12 kg. For the mass of the robot body, see the specifications.

1.2.2 Transporting the Robot

When transporting the robot, observe the following procedures, fully confirming the cautions.

! DANGER

- When transporting the robot, be sure to use the attached lifting fixtures.
- Be sure to secure the arm with the attached clamp before transporting the robot. Otherwise, a hazardous situation may be caused because the arm will move when the robot is suspended.

When transporting the robot by a crane, observe the following procedures.

1) Make sure that the arm clamps are attached.

2) Remove the hexagon bolts and washers, then attach the M10 eyebolt to each screw hole where the lifting fixture is set.

3) Pass wire ropes through the eyebolts. To assure safety, be sure to use four (4) wire ropes. If there is a fear that the wire rope will contact the robot body due to the length of the wire rope, use a cushion.

4) Lift up the robot carefully, using the crane.

5) After installation, remove the clamps, lifting fixtures and eyebolts.

6) Set the attached hexagon bolts and washers to the four (4) screw holes.

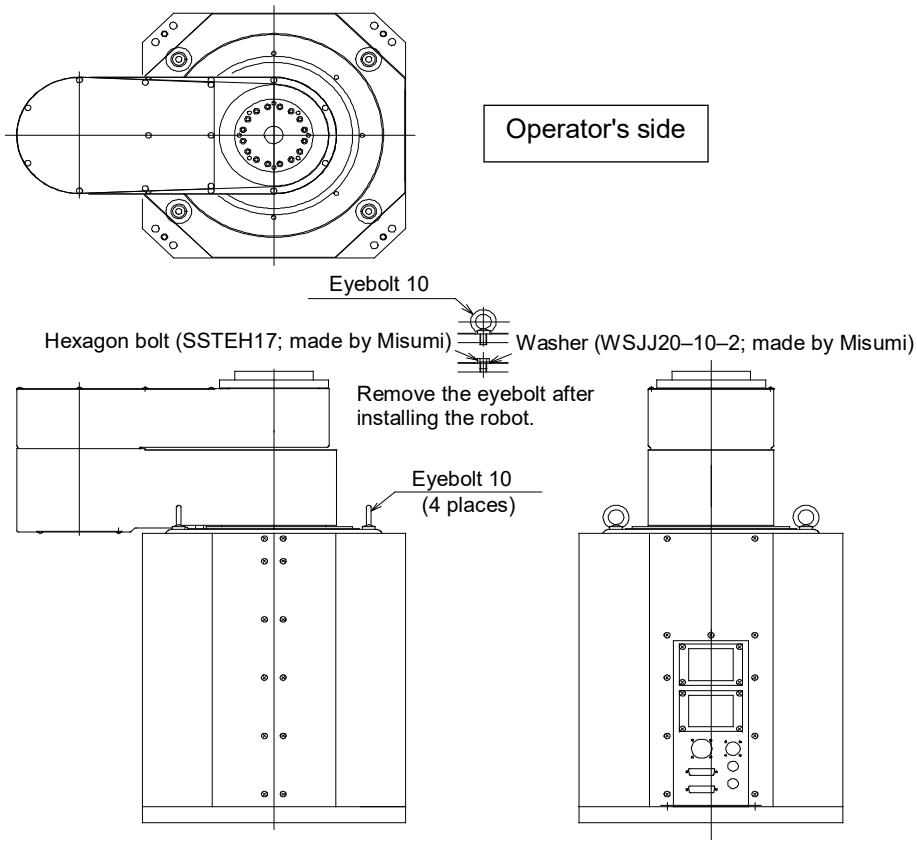


Fig. 1.2 Robot lifting posture

! CAUTION

- Use the wires which can well withstand the robot mass.
- The robot will tilt more or less when lifted. Carefully lift it up.
- When raising and lowering the robot, take careful precautions not to exert any excessive impact (9.8 m/s^2 or over) on the robot.
- When placing the robot on the floor, etc., take utmost care not to have your hands or feet caught under the robot.

1.2.3 Transporting the Controller

Before transporting the controller, disconnect all cables, teach pendant and FG unit.

DANGER

- When placing the controller on the floor, etc., take utmost care not to have your hands or feet caught under the controller.

1.3 Storage

Both the robot and controller should not be stored for a long period of time after unpacking them. When this is unavoidable, however, be sure to observe the following cautions on storage.

1.3.1 Cautions on Robot Storage

CAUTION

- Secure the base completely to prevent the robot from falling over.
- Keep the robot out of direct sunlight. Otherwise, the parts may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.
- When the robot motor drive cable was disconnected from the controller, connect the short-circuit connector to the same cable.

1.3.2 Cautions on Robot Controller Storage

CAUTION

- Keep the robot controller out of direct sunlight. Otherwise, the controller interior will be heated up, resulting in faults.

Section 2 Installation

2.1 Installation Environment

Table 2.1 shows the environmental conditions for the location in which the robot and controller TS2000 are to be installed.

Table 2.1 Environmental conditions for the robot and controller

Item	Specifications
Temperature	At operation : 0 ~ 40°C At storage : –10°C ~ 50°C
Humidity	20 ~ 90 % (non-condensing) DO NOT install the robot and controller at places where they are exposed to splash of liquids such as water.
Altitude	1,000 m or less
Vibration	At operation : 0.98 m/s^2 or over At transport : 9.8 m/s^2 or less
Dust	No inductive dust should be involved. Consult with SHIBAURA MACHINE first if you wish to use the robot and controller in a hostile environment.
Gas	No corrosive gas or combustible gas should be involved.
Sunlight	The robot and controller should not be exposed to direct sunlight.
Power source noise	No device causing an excessive noise should exist nearby.
Field	No device causing a strong field should exist nearby.

! DANGER

- DO NOT place the robot and controller near combustible. Otherwise, fires could be caused if the material ignites due to a fault, etc.

2.2 Installing the Robot

Before actually installing the robot, you should plan a layout and take into consideration such matters as working envelope, coordinate system and space for maintenance.

2.2.1 External Dimensions

See the external view drawing of the robot attached to the specifications.

2.2.2 Working Envelope

See the external view drawing of the robot attached to the specifications.

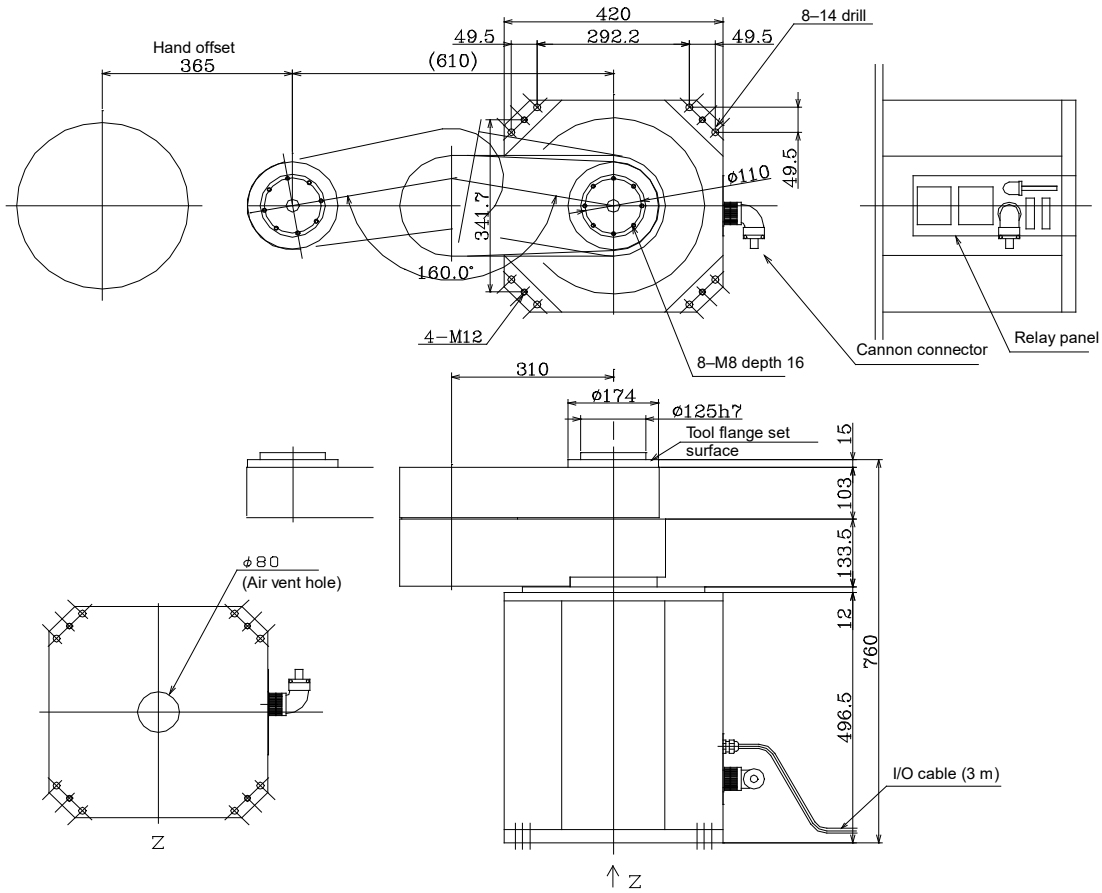


Fig. 2.1 External view of SR–624HC (Robot with loading capacity of up to 35 kg)

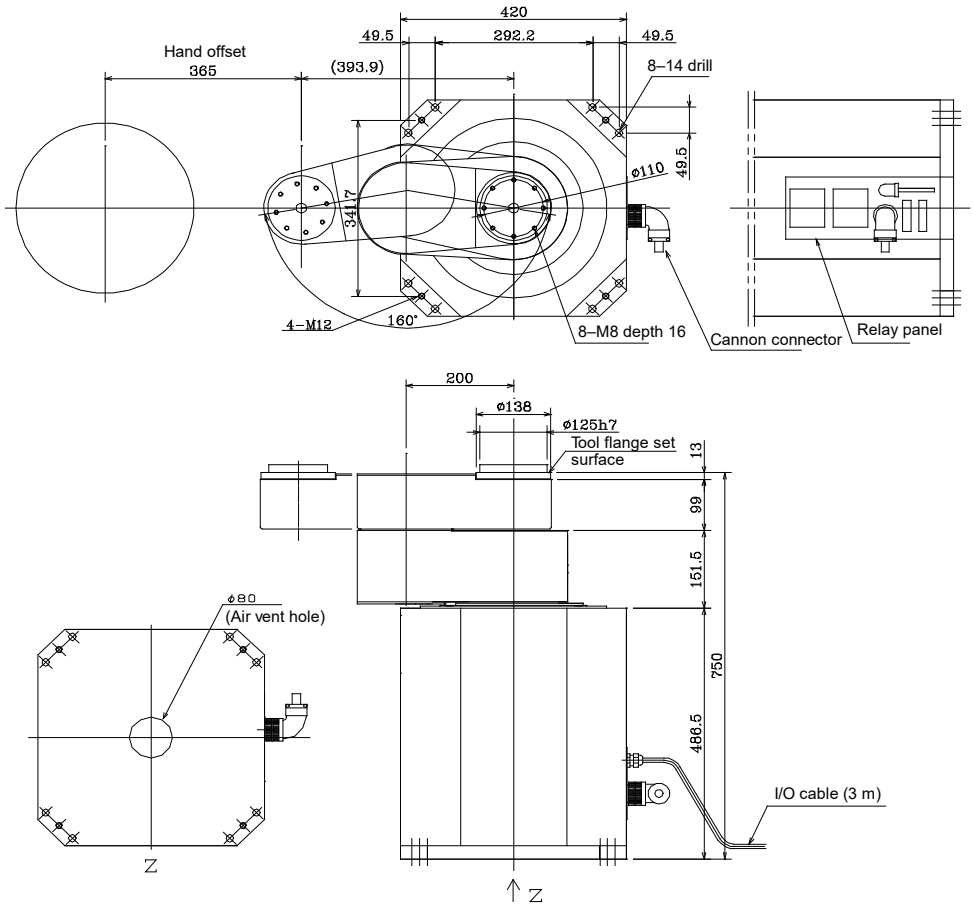


Fig. 2.1a External view of SR–404HC (Robot with loading capacity of up to 20 kg)

2.2.3 Robot's Origin Posture and Coordinate System

The origin posture and coordinate system of the robot are exemplified in Fig. 2.2.

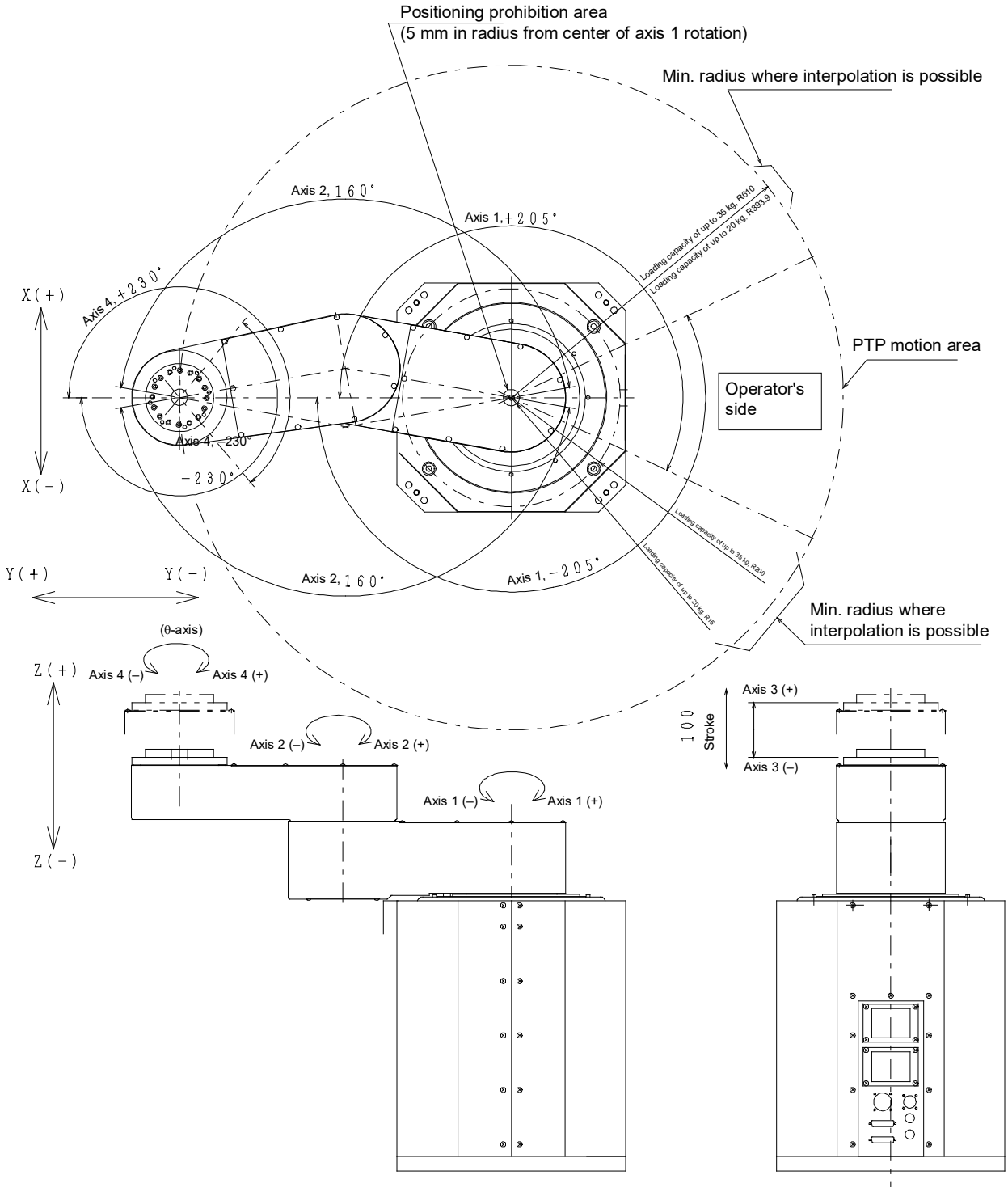


Fig. 2.2 Coordinate system of the robot

2.2.4 Robot Installation Procedures

The robot is secured, using the set holes of the robot base. The robot base set areas as shown in Fig. 2.3 should be rigid enough and their surfaces should be machined with flatness accuracy as small as 0.2 or less.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing the robot on the frame, make sure that the frame has enough rigidity. If the robot is installed on a frame that is not rigid enough, vibration will be caused during the robot operation, resulting in faults.
  When installing the robot on the floor, secure it with anchor bolts, etc.
- Install the robot on the leveled place. Otherwise, the robot performance will deteriorate or the robot will go wrong.

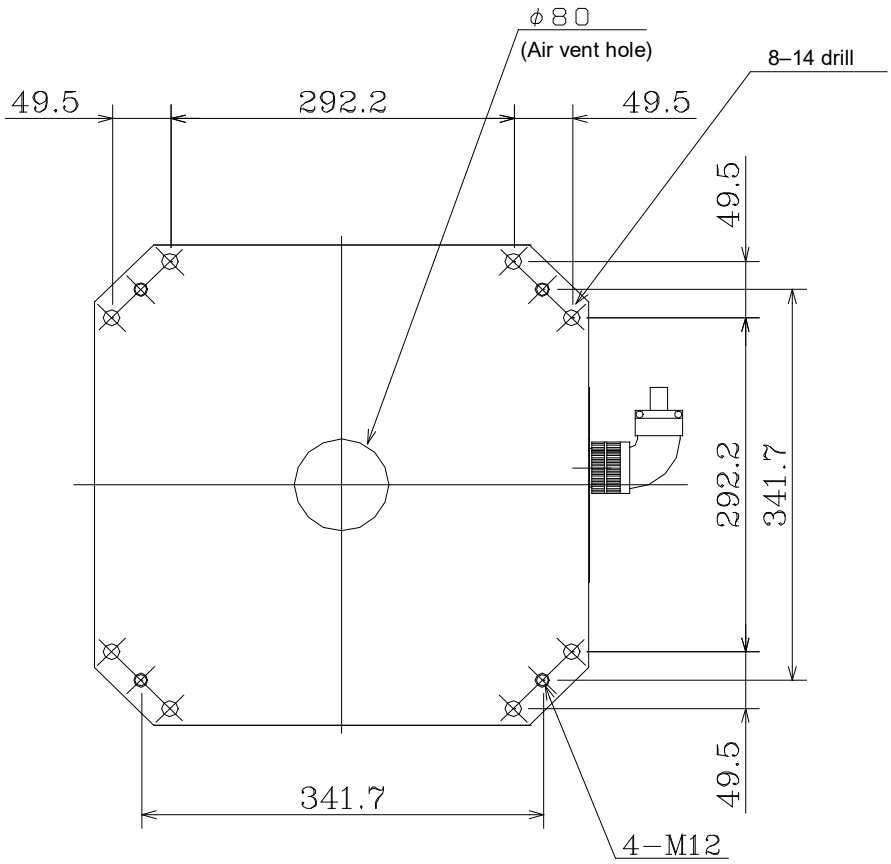
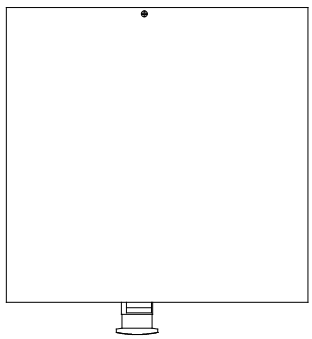


Fig. 2.3 Robot base

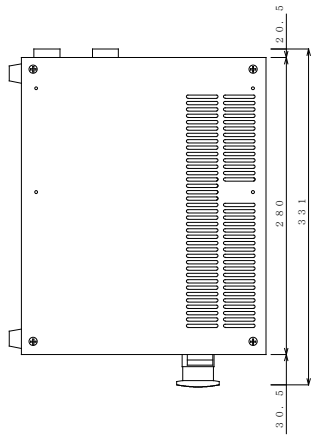
2.3 Installing the Controller

2.3.1 External Dimensions

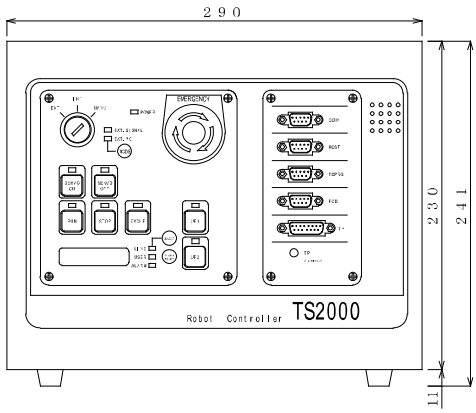
Fig. 2.4 shows the external dimensions of the robot controller.



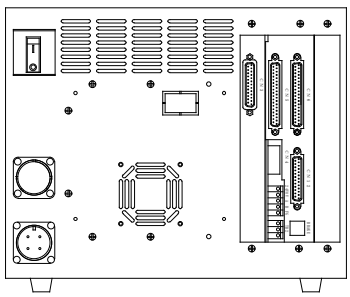
Top view



Right view



Front view



Rear view

Fig. 2.4 External view of TS2000 controller

2.3.2 Rack Mounting Dimension

When instaling the TS2000 robot controller in the rack, set the side brackets, using the screw holes on both ends of the front panel, then secure the controller to the rack.
(Side brackets [1] in Fig. 2.5 are optional.)

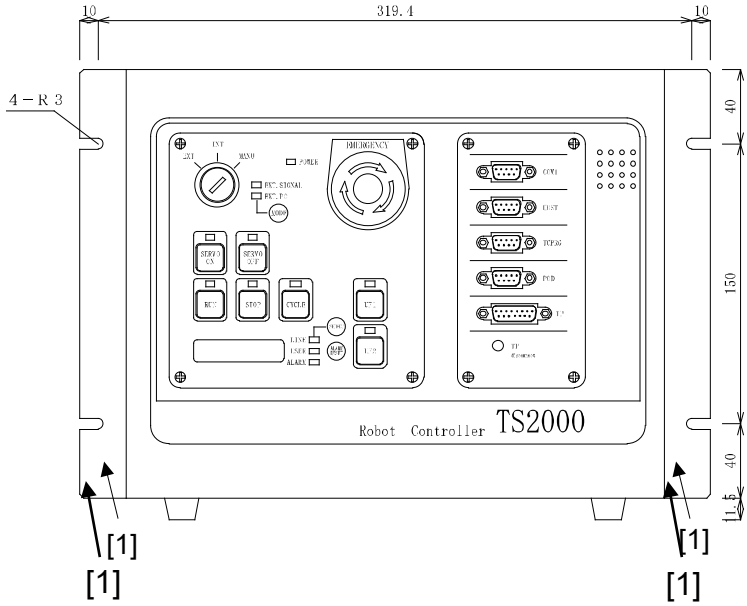


Fig. 2.5 Dimensions of controller rack clamp screw holes

2.3.3 Cautions on Rack Mounting

Be careful of the following matters when rack-mounting the TS2000 robot controller.

a) When mounting the controller in the rack, use the screw holes provided on both ends of the front panel and secure the controller. (The optional side brackets are required.)

! CAUTION

- If the rack is completely sealed, provide holes for air vent, or compulsively ventilate the rack with a fan or cool the rack indirectly, so that the heat will not be trapped in the rack.
  If the heat is trapped in the rack or controller, the controller will malfunction.

b) As the cable connectors are connected to the rear side of the controller, keep a space of 110 mm on the rear side. To perform maintenance of the controller, remove the upper cover. (See Fig. 2.6.)
When installing the controller, fully consider the controller maintenance. When the controller is housed in the rack, it should be removed from the rack at maintenance. Specifically, be careful of the following points.

1) Arrange the cables around the rear side of the controller (so that the controller can be removed).

2) Arrange the cables between the controller and control panel when the control panel is separated (so that the controller can be removed).

3) Connect all cables in a position where the robot can be operated even if the controller is removed from the rack.

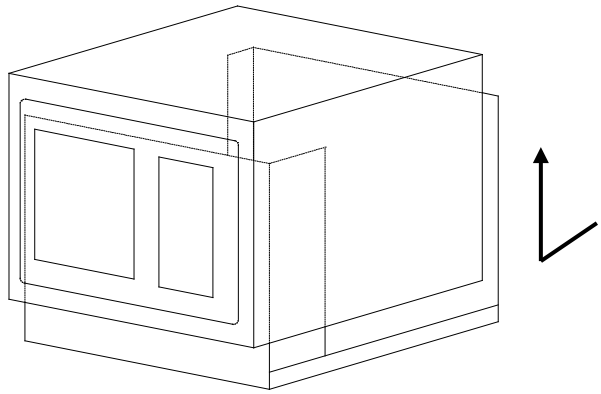


Fig. 2.6 Removing upper cover

c) When the controller is assembled in the rack, the weight of the controller should be supported by the legs of the controller. The rack mount screw holes of the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with these screws.

d) On the front side of the controller, a clearance of about 90 mm should be kept for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of about 50 mm is required for connecting a dummy plug.

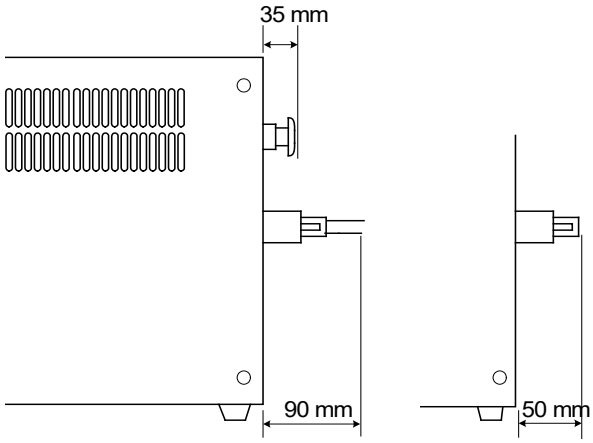


Fig. 2.7 Clearance of controller front panel

2.3.4 Cautions on Direct Installation

When two (2) or more controllers are used, keep a clearance of 50 mm or over in the horizontal direction and a clearance of 100 mm or over in the vertical direction between the controllers.

! CAUTION

- Keep a ventilation space on the lateral side of the controller to prevent stoppage of the air vent holes. Also, keep the space equal to the length of the support below the bottom surface. Otherwise, the cooling ability will drop, causing faults.
- DO NOT stack the controllers. DO NOT put objects on the top of each controller.

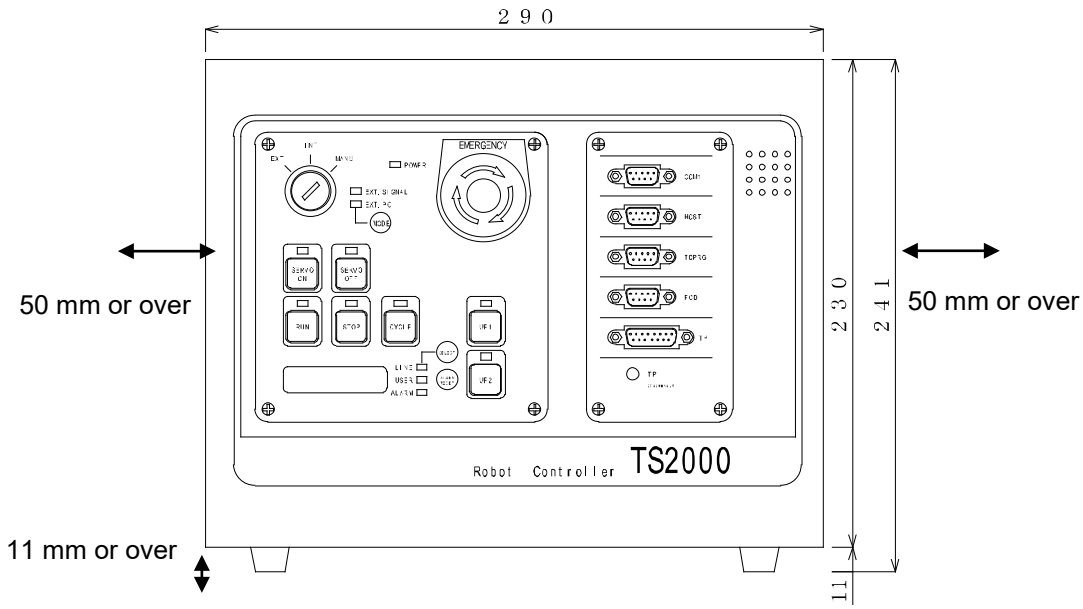


Fig. 2.8 Controller ventilation space

2.4 Cautions on Handling of Teach Pendant

When handling the teach pendant, observe the following precautions.

! CAUTION

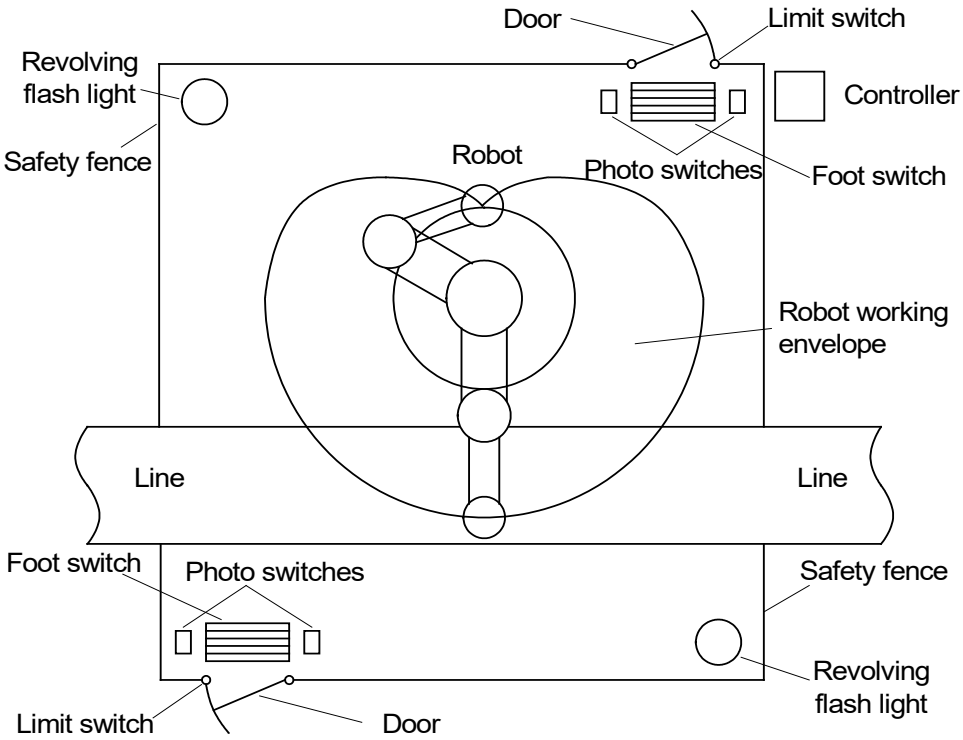
- DO NOT drop the teach pendant or hit it against anything.
- DO NOT pull the cable coming out of the teach pendant.
- DO NOT press the switches on the teach pendant with a sharp cutter, pencil or ball-point pen.
- DO NOT place or use the teach pendant near open flames.
- DO NOT leave the teach pendant in direct sunlight for a long period of time.

2.5 Safety Measures

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide a safety fence to prevent anyone from entering the zone easily. The hazard zone is the zone near the working envelope of the robot, where a hazardous state could occur if a person enters.

c) Mount the limit switches, photo switches, foot switches, etc., on the entrance and exit doors of the safety fence and provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent normal close contact with compulsive opening function and must not be recovered automatically.



d) The controller should be installed at a place outside the hazard zone where the operator can overlook the robot motions.

2.6 Position Detector Backup Batteries

The robot has batteries to back up positional information on position detectors in its base. Unless the robot is used for a long period of time, the battery voltage will drop and the positional data of zero point will be lost.
Also, if the battery voltage for backup of the X8HC printed board SRAM of controller has dropped, absolute position data set by the user will be lost. To avoid this, replace each battery on a regular basis.

If the battery line voltage has dropped, an alarm of level 1 as shown below occurs. When this happens, replace the battery immediately.

Error code	Descriptions
1–049	Axis1 Battery alarm
1–065	Axis2 Battery alarm
1–081	Axis3 Battery alarm
1–097	Axis4 Battery alarm
1–113	Axis5 Battery alarm
1–129	Axis6 Battery alarm
1–145	MAIN Battery alarm

If the line voltage has dropped, an alarm of level 8 as shown below occurs. When this happens, replace the battery with a new one and adjust the zero point of the robot, if shifted, because there is a fear that the position encoder backup data has been lost.

Error code	Descriptions
8–065	Axis1 Encoder abnormal
8–097	Axis2 Encoder abnormal
8–129	Axis3 Encoder abnormal
8–161	Axis4 Encoder abnormal
8–193	Axis5 Encoder abnormal
8–225	Axis6 Encoder abnormal

Section 3 System Connections

3.1 Cable Wiring

This section describes the various types of cables and connectors and explains how these are to be hooked up.

3.1.1 Connector Arrangement on the Controller

The cables connected to the SR7000 robot controller are shown in Fig. 3.1

[1] Power cable
[2] Motor cable
[3] Encoder cable
[4] Robot control signal (HAND i/o)
[5] External I/O signal
[6] Brake signal cable
[7] Distribution I/O signal cable

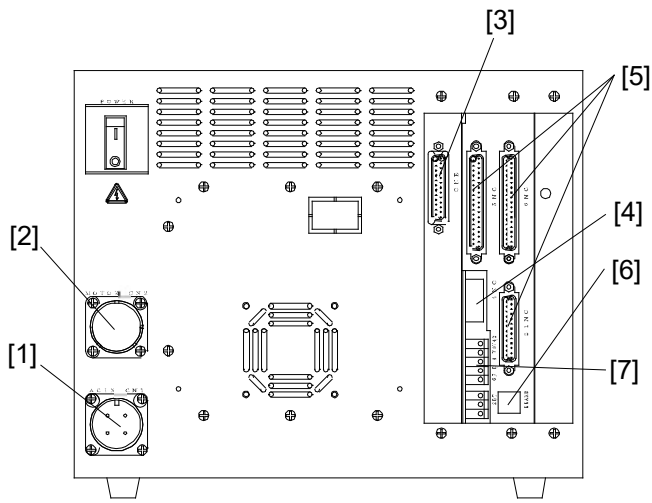


Fig. 3.1 Connector arrangement on the controller

In this paragraph, we explain how to connect cables [1] ~ [4]. For information on how to connect cables [5], [6] and [7], refer to the Interface Manual.

3.1.2 Connecting Power Cable "CN1" (Fig. 3.1–[1]; Plug connector included)

The power cable is used to supply the main power to the controller.

Table 3.1 Power supply specifications

Item	Specifications
Power supply	Single phase, AC 180 ~ 250 V, 50/60 Hz±1 Hz
Power supply capacity	1.7 kVA (SR–624HC, SR–404HC)
Instantaneous power failure	40 msec or less
Grounding	Grounding with grounding resistance of 100 Ω or less

The connector is CN1 (Fig. 3.1–[1])
As the cable is not attached, the user should fabricate it, using the plug connector connected to CN1 on the controller side.

CN1 plug connector	Type: JL04V–6A18–10SE–EB	Maker: Japan Aviation Electronics Industry
CN1 cable clamp	Type: JL04–18CK	Maker: Japan Aviation Electronics Industry
Wire	3.5 mm^2 ~ 5.5 mm^2	

The connector (JL04V–6A18–10SE–EB) is soldered to the wires.

! DANGER

- Be sure to use the designated wires. Otherwise, fires or faults could be caused.
- When connecting the connector and wires, make sure of the terminal arrangement.
- After the connection, confirm the connection, using a tester, etc.

For the terminal arrangement, see Para. 3.1.7.

! CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, "8–027 Slow Charge error" occurs.
  When this happens, make sure that the line voltage at the controller power connector satisfies the specified input power of the controller, and that the same line voltage is stabilized.
- For details of the 8–027 error, see the Operator's Manual.

3.1.3 Connecting Motor Cable "CN2" (Fig. 3.1–[2]; Cable included)

The motor cable connects the controller and robot, and supplies the power required to rotate the motor from the controller servo driver to each axis motor of the robot. The motor cable connector is CN2 (Fig. 3.1–[2]).

3.1.4 Connecting Encoder Cable "CN3" (Fig. 3.1–[3]; Cable included)

The encoder cable is a signal line used to transmit data from each axis encoder (i.e., device used to detect the revolution (and thereby the position) of each axis) to the controller. The motor cable connector is CN3 (Fig. 3.1–[3]).

3.1.5 Connecting Robot Control Signal Cable "CN4" (Fig. 3.1–[4]; Cable included)

The robot control signal cable is used for the brake ON/OFF signals for locking the motor shaft, hand control signals and others.
(Input and output of robot control signals are not possible for this model.)
The motor cable connector is CN4 (Fig. 3.1–[4]).

3.1.6 Inserting and Removing Connectors

! CAUTION

- Before inserting or removing any controller cable, be sure to turn off the POWER switch.
- When disconnecting a cable, be sure to pull the plug in lieu of the cable. Otherwise, you may break the cable.

a) Circular connectors: CN1, CN2

Completely push the connector on the cable side into the connector on the controller side. Then turn CW and tighten the cable side lock screw. If it is loosened, an accident due to contact failure of connector will occur.

To remove the connector, turn CCW and loosen the lock screw, then draw out the cable side connector.

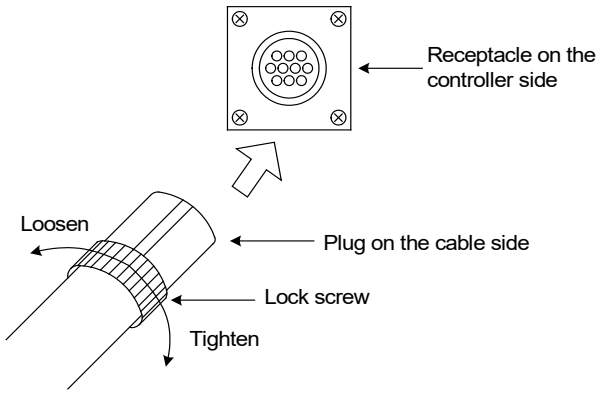


Fig. 3.2 Inserting and removing circular connector

b) Rectangular connectors: CN3, CN4, CN5, CN6, CN12

Completely push the connector on the cable side into the connector on the controller side. Then tighten the lock screws on both ends of the cable side connector with a screwdriver. If they are loosened, an accident due to contact failure of connector will occur.

To remove the connector, loosen the lock screws, then draw out the cable side connector. CN4 is the quick-operated lock type connector.

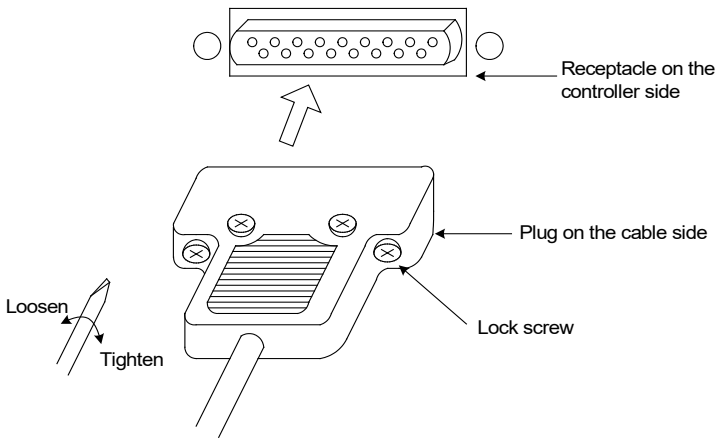
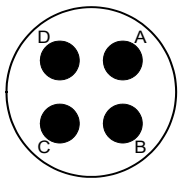


Fig. 3.3 Inserting and removing rectangular connector

3.1.7 List of Connector Terminal Arrangement Examples

a) Power cable connector CN1



Connects to controller.
Type: JL04V–2E18–10PE–B
Maker: Japan Aviation Electronics Industry

A
B — Single phase, AC 180 ~ 250 , 50/60 Hz

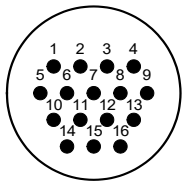
C

D — Grounding (Grounding with grounding resistance of 100 Ω or less)

! DANGER

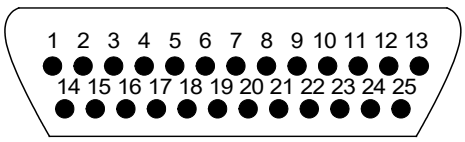
- Completely connect the grounding cable.
  Otherwise, electric shocks or fires could be caused if a fault or leak current occurs, in addition to malfunction by noise.

b) Motor cable connector CN2



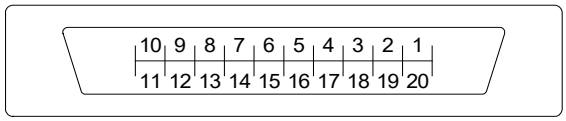
Connects to controller.
Type: JR25RK–16S
Maker: Hirose Electric

c) Encoder cable connector CN3



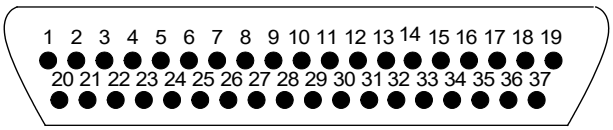
Connects to controller.
Type: XM2A–2501
Maker: OMRON

d) Robot control signal cable connector CN4



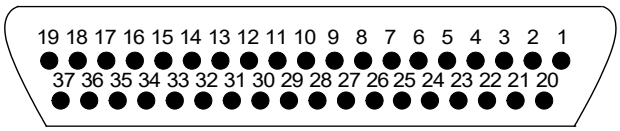
Connects to controller.
Type: PCR–E20LMD
Maker: Honda Tsushin Kogyo

e) External input signal cable connector CN5



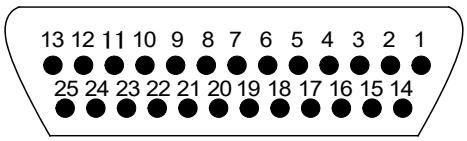
Connects to controller.
Type: XM2C–3712–112
Maker: OMRON

f) External output signal cable connector CN6



Connects to controller.
Type: XM3B–3722–112
Maker: OMRON

g) External I/O signal cable connector CN12



Connects to controller.
Type: XM3B–2522–112
Maker: OMRON

3.2 Controller Connector Signals

3.2.1 Connector Signal Connection Diagrams

The diagrams showing which signals correspond to which terminals are shown in Section 2 of the Interface Manual.

3.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals	CN5–16	(STOP)
	CN6–34	(SVOFF)
	CN5–17	(BREAK)
	CN6–16, 35	(EMS1B ~ EMS1C)
	CN6–17, 36	(EMS2B ~ EMS2C)

These signals are already jumpered for the connectors provided for the TS2000 controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side CN5, CN6 connectors.
Also, if the following signals are not used as the system signals, be sure to jumper them.

CN5–36 (LOW_SPD)
CN5–35 (CYCLE)

Connector jumpers

CN5		CN6	
16–18	17–18	18–34	—
(35–37)	(36–19)	16–35	17–36

! CAUTION

1. If the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
2. Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
3. Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
4. Unless the STOP signal is jumpered, automatic operation of the robot is not possible.

3.3 Separating Control Panel from Controller

3.3.1 Removing Control Panel

Remove the control panel in the following manner.

a) Loosen the four (4) screws at the four (4) corners, which secure the control panel.

b) Remove these screws, then carefully draw out the control panel toward your side. (Be careful of the cable connected on the rear side.)

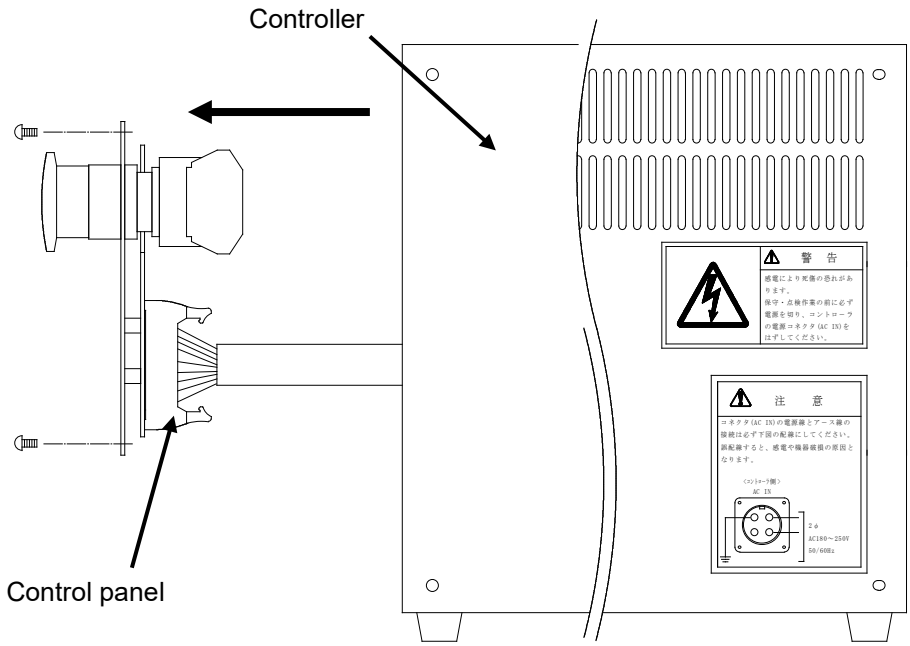


Fig. 3.4 Removing control panel

3.3.2 Cable between Controller and Control Panel

The cable required to connect the control panel and controller when they are installed separately can be provided optionally.

3.3.3 Control Panel Mounting Dimensions

The dimensions of mounting the control panel are shown in Fig. 3.5.
Cross truss head screws (ø3×6, ZN3–B) are used.

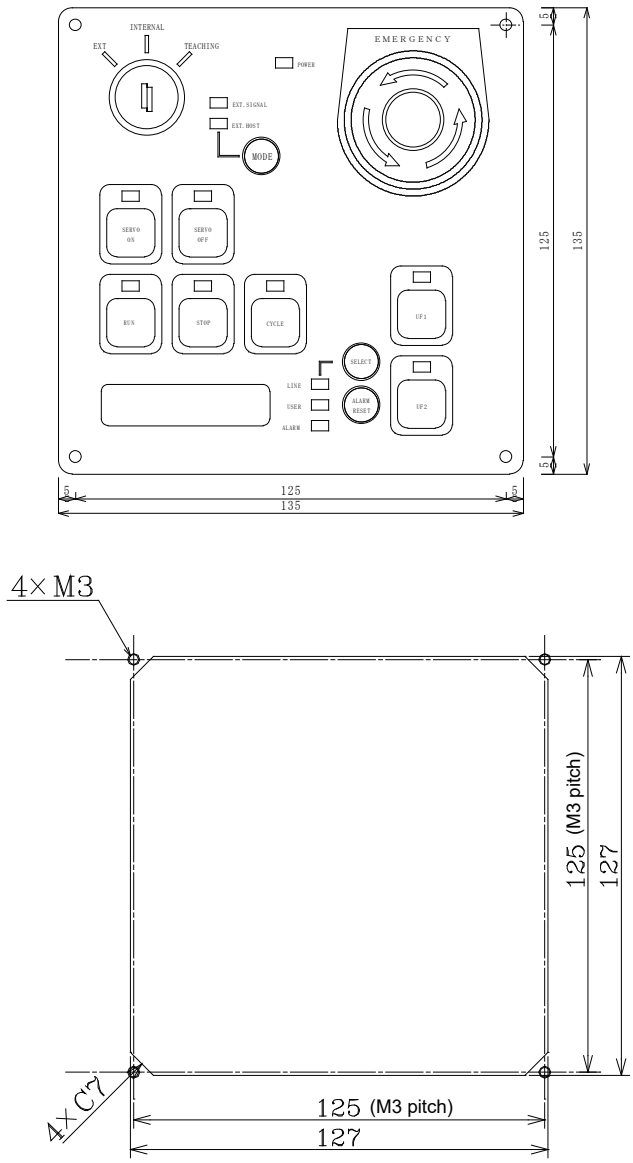


Fig. 3.5 Control panel mounting dimensions

3.3.4 Mounting Dummy Panel on Controller

When the control panel has been disengaged from the controller, mount a dummy panel on the place where the control panel was set before. The dummy panel, set parts, etc. are provided optionally.

a) Connect the cable connector which was disconnected when separating the controller from the control panel, to the rear side of the dummy panel, then screw both ends of the connector. When mounting the connector, use the cross truss screws (ø3×6, ZN3–B).

b) Screw the dummy panel into the controller.

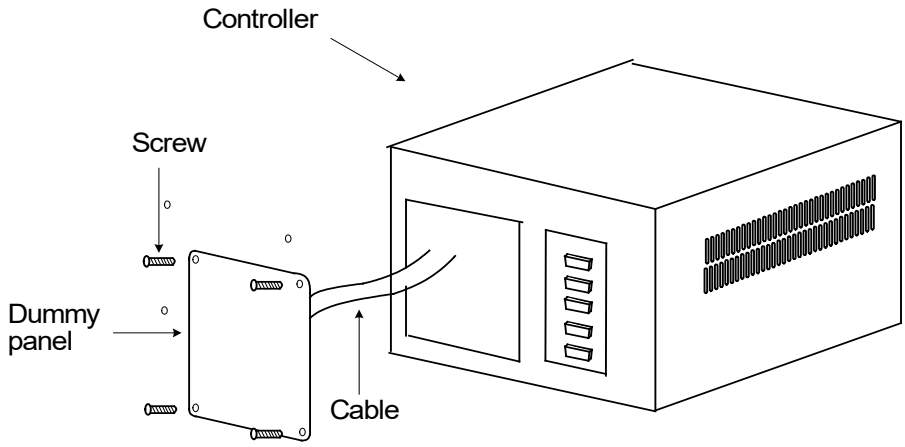


Fig. 3.6 Mounting dummy panel

3.3.5 Dimensions when Separating Control Panel

Fig. 3.7 shows the connections of the control panel and dummy panel. Provide a clearance of about 50.6 mm on the rear side of the separated control panel. When the cable is connected to the dummy panel of the controller, provide a clearance of 80 mm or so in front of the controller as the cable connector sticks out of the panel surface.

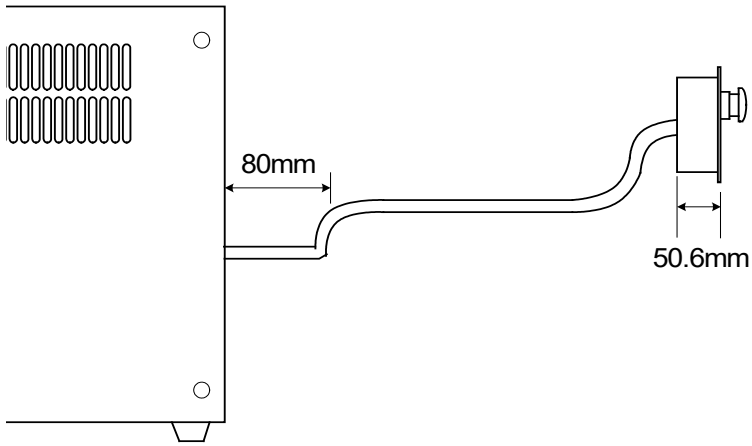


Fig. 3.7 Cable connections of dummy panel and control panel

Section 4 Hand Interface

4.1 Hand Wiring

The hand wiring is shown in Fig. 4.1, which is used for the input signals of sensors and control signals of solenoid valves.

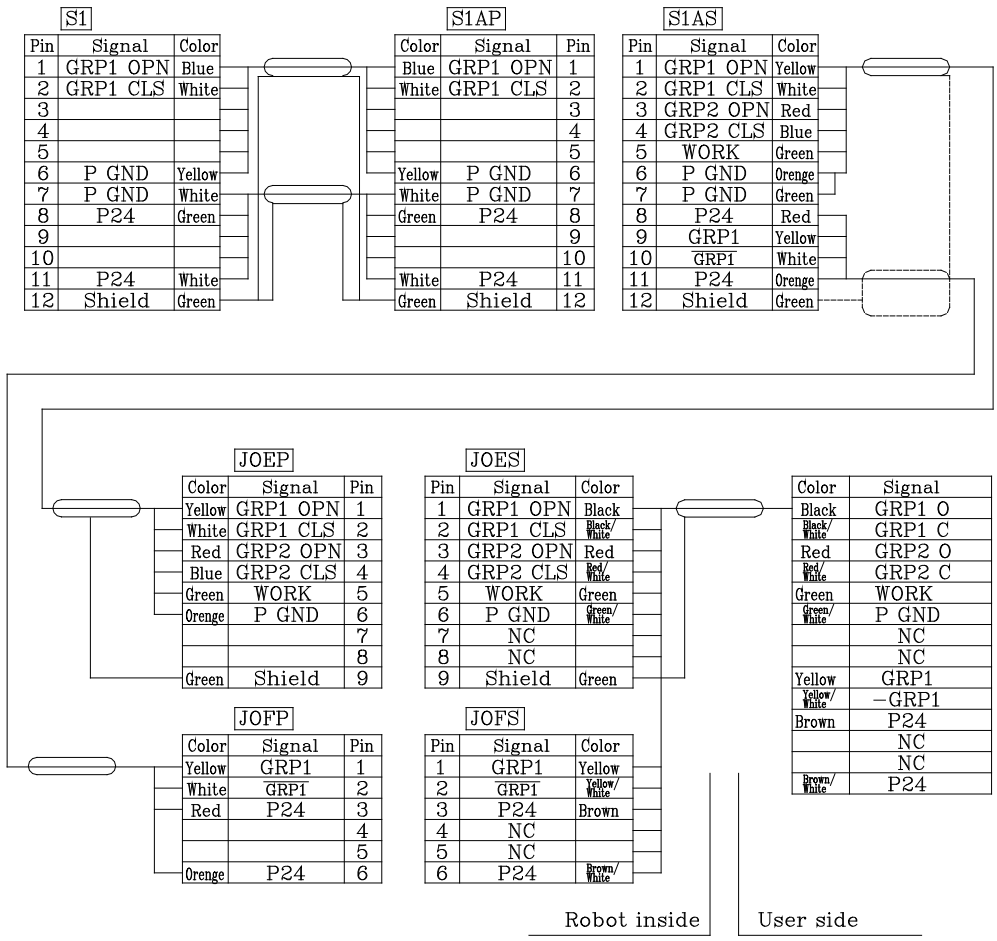


Fig. 4.1 Hand wiring

CAUTION

- Always use a highly flexible robot cable. It should be secured below the arm with a cable clamp, etc. Unless a robot cable is used, the wire may be broken.